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Exploring learners' self-regulated learning behaviors in generative AI-assisted academic writing: a cluster and sequential analysis

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ABSTRACT

Despite the growing integration of generative artificial intelligence (GenAI) into academic writing, learners' self-regulated learning (SRL) behavior patterns and their relationships with individual characteristics remain insufficiently understood. Drawing on 37 hours of screen-recorded GenAI-assisted writing data, comprising 3268 behavioral units from 30 graduate students, the present study examines the dynamic SRL behaviors of students across different learner characteristic clusters in GenAI-assisted academic writing. This study employs multiple learning analytics methods (clustering, descriptive statistics, non-parametric tests, and lag sequential analysis) and incorporates English proficiency and final writing performance as key learner characteristics. The results show that (a) three student clusters emerge based on English proficiency and final writing performance: the Higher-English-Proficiency–Lower-Writing-Performance group, the Higher-English-Proficiency–Higher-Writing-Performance group, and the Lower-English-Proficiency–Higher-Writing-Performance group; (b) higher-performing groups exhibit more cyclical and well-coordinated SRL behavioral patterns, whereas the lower-performing group demonstrates relatively less coordinated SRL behavioral patterns; and (c) among higher-performing students, those with lower English proficiency tend to engage more extensively in SRL behaviors and achieve comparable writing performance; however, this pattern also raises concerns about a possible overreliance on GenAI. Based on these findings, a series of theoretical and pedagogical implications are proposed to support the enhancement of students' SRL in GenAI-assisted academic writing contexts.

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KEYWORDS
Academic writing process; self-regulated learning; lag sequential analysis; generative artificial intelligence

1. Introduction

Academic writing is widely recognized as a core competency in higher education, reflecting learners' academic literacy as well as their ability to construct and communicate disciplinary knowledge (Bacha, 2002; Swales & Feak, 2012). For English as a Foreign Language (EFL) learners, however, academic writing remains one of the most challenging language skills due to its complex cognitive, linguistic, and rhetorical demands (Al Fadda, 2012; Lin & Morrison, 2021). Prior research has consistently highlighted the critical role of English proficiency in shaping writing outcomes. Learners with stronger linguistic foundations generally produce higher-quality texts, whereas those with weaker proficiency often struggle to express complex ideas effectively (Lahuerta Martínez, 2024; Sasaki & Hirose, 1996; Wu et al., 2021). Such patterns are consistent with the Matthew effect (Stanovich, 2009), whereby initial advantages in language proficiency accumulate over time and amplify performance differences among learners.

The rapid emergence of generative artificial intelligence (GenAI) has introduced new possibilities for academic writing support. Tools such as ChatGPT provide real-time assistance with idea generation, language refinement, and revision (Kim et al., 2025; Koltovskaia et al., 2024; Su et al., 2023; Yang & Li, 2024). Through such linguistic scaffolding, GenAI may reduce learners' dependence on their existing language resources, allowing them to focus more on higher-order aspects of writing. As a result, the traditional relationship between English proficiency and writing performance may become less deterministic in GenAI-assisted writing environments.

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From a compensatory perspective, learners' self-regulated learning (SRL) may play an increasingly important role in such environments. SRL refers to learners' ability to plan, monitor, and evaluate their learning activities in order to achieve academic goals (Zimmerman, 1990; Zimmerman, 2002). A growing body of research has demonstrated the positive impact of SRL strategies on writing performance (Teng & Zhang, 2024; Zhu et al., 2024). In particular, Teng and Zhang (2024) found that English proficiency moderated the relationship between self-regulated writing strategies and writing outcomes. However, most existing studies have relied on self-reported SRL measures and have primarily examined these relationships at the outcome level (e.g. Teng & Zhang, 2024; Zhu et al., 2024), leaving limited understanding of how learners enact observable self-regulatory behaviors during the writing process. Moreover, much of this research has been conducted in conventional writing contexts (e.g. Aksela et al., 2025; Teng & Zhang, 2024; Zhu et al., 2024), providing limited insight into how SRL operates in emerging GenAI-assisted writing environments.

These limitations become particularly salient in GenAI-assisted writing contexts. Unlike traditional feedback settings, GenAI tools provide continuous, interactive support throughout the writing process, requiring learners to strategically regulate when and how they engage with AI-generated suggestions. Emerging empirical studies indicate that learners using GenAI-based writing support tend to engage more frequently in planning, evaluation, and revision activities than in traditional writing environments (Fan et al., 2025; Teng et al., 2026). At the same time, the effectiveness of such AI-assisted writing support may vary across learners with different proficiency levels (Teng et al., 2026). These findings suggest that writing performance in GenAI-assisted contexts may depend not only on learners' English proficiency but also on how effectively they regulate their interactions with AI-generated support during the writing process. Yet it remains unclear how these SRL behaviors unfold during the writing process and how they relate to learners' individual characteristics in GenAI-assisted contexts. In particular, if GenAI reduces language-related constraints, the relationship between English proficiency and writing performance may manifest in multiple configurations. Rather than assuming a single linear relationship between proficiency and performance, it is therefore important to explore whether distinct learner profiles emerge in GenAI-assisted writing contexts. For example, some learners with lower proficiency may achieve relatively high writing performance through effective use of AI support, whereas others with higher proficiency may not necessarily achieve comparable outcomes. Examining such proficiency–performance configurations can provide insights into how learners regulate their writing processes when interacting with GenAI. To address this issue, the present study employs clustering and lag sequential analysis to examine how students with different proficiency–performance configurations enact SRL behaviors during GenAI-assisted academic writing. Accordingly, this study addresses the following research questions:

RQ1: What clusters emerge based on students' prior English proficiency and final writing performance in GenAI-assisted EFL academic writing?

RQ2: How do SRL behavior frequencies vary across clusters during GenAI-assisted academic writing?

RQ3: How do the SRL behavioral patterns differ across clusters during GenAI-assisted academic writing?

2. Theoretical framework: self-regulated learning theory

Self-regulation is a critical factor in achieving academic success (Zimmerman, 1990). Self-regulated learners view learning as a systematic and controllable process, actively seeking information when needed and adopting effective strategies to master the required knowledge and skills (Borkowski et al., 1990; Pintrich, 2000; Zimmerman, 1990).

Various SRL theoretical models provide valuable insights into how learners regulate their behaviors throughout the learning process (e.g. Pintrich, 2000; Winne & Perry, 2000; Zimmerman, 2002). Among these, Zimmerman's (2002) cyclical model of self-regulation provides a particularly clear and concise framework. Previous studies have operationalized Zimmerman's model to examine learners' strategic behaviors in various contexts, such as gamified EFL reading context (Maimaiti & Hew, 2025) and human-AI collaborative programming learning environment (Ma et al., 2026), confirming the model's explanatory power in capturing dynamic self-regulatory patterns.

According to Zimmerman's (2002) model of self-regulation, learners' self-regulatory behaviors unfold across three phases: forethought phase, performance phase, and self-reflection phase. The forethought phase involves cognitive processes and motivational beliefs that occur prior to learning, encompassing activities such as goal setting and strategic planning. The performance phase refers to the process in which students engage in the behavioral implementation of their learning strategies. The self-reflection phase refers to the stage in which learners evaluate and reflect on their performance after completing a learning task or activity. It primarily focuses on how learners review their learning outcomes and make improvements. Importantly, Zimmerman (2002) emphasized that self-regulation is not a linear process but a dynamic and cyclical system in which the processes of forethought, performance, and self-reflection are closely interconnected and continuously influence one another (Zimmerman, 2002; Zimmerman & Kitsantas, 1999; Zimmerman & Moylan, 2009).

In the context of EFL learning, students need to employ a range of SRL strategies tailored to their goals. These include metacognitive strategies, such as planning, monitoring, and evaluating language learning activities; cognitive strategies, which involve targeted practice and processing of language knowledge and skills, such as resourcing, reasoning, summarizing, and translation; and social strategies, such as seeking assistance from others when encountering difficulties (Cohen, 2014). Zimmerman's cyclical model provides a useful theoretical foundation for explaining these strategic choices, as it highlights the cyclical nature of learning processes and clarifies how learners regulate their behaviors across different phases of learning, such as forethought, performance, and self-reflection.

3. Literature review

3.1. GenAI-assisted academic writing

Academic writing is widely recognized as a crucial skill for academic success (Bacha, 2002; Lin & Morrison, 2021; Swales & Feak, 2012). However, academic writing can be particularly challenging for students who have not developed sufficient linguistic competence and content knowledge (Lin & Morrison, 2021; Qin & Karabacak, 2010). GenAI, with its outstanding performance in areas such as text generation, language refinement, and idea expansion, has become a powerful tool for assisting academic writing (Kim et al., 2025; Koltovskaia et al., 2024; Su et al., 2023), and has sparked widespread discussions among scholars.

Current research on GenAI-assisted academic writing primarily examines individuals' perceptions of using GenAI tools for academic writing, describing their advantages and potential risks (e.g. Kim et al., 2025; Kohnke, 2024; Yan, 2023). This line of research mainly relies on subjective measures, such as self-reported questionnaires, interviews, or reflective journals. Another line of research has investigated the effectiveness of GenAI-assisted academic writing, typically adopting experimental or quasi-experimental research designs to evaluate students' learning outcomes (e.g. Alsofyani & Barzanji, 2025; Hong & Shin, 2025).

Beyond these perception- and outcome-oriented studies, an increasing body of research has begun to analyze process-based data to uncover the dynamic interactions between GenAI and students during the writing process (e.g. Koltovskaia et al., 2024; Nguyen et al., 2024). For instance, Nguyen et al. (2024) investigated doctoral students' writing behavior patterns when collaborating with GenAI. Koltovskaia et al. (2024) used screen recordings to explore students' behavioral engagement in GenAI-assisted academic writing. Their research provided valuable insights into the dynamics of human-AI interactions in academic writing. However, these studies focused primarily on participants' specific actions (e.g. copy-pasting, adopting GenAI feedback) and largely neglected the strategic aspects of writing, particularly SRL behaviors, leaving an important aspect of the writing process underexplored.

3.2. SRL in EFL writing contexts

The regulatory role of SRL behaviors in EFL writing has received particular attention (Sun & Wang, 2020; Teng & Zhan, 2023; Teng & Zhang, 2024; Zhu et al., 2024). Some studies have identified and categorized SRL behaviors in the writing process. For instance, Teng and Zhang (2024) conducted an empirical study to examine the effects of SRL strategies and L2 English proficiency on multimedia writing performance, categorizing SRL strategies into goal setting, strategic planning, elaboration, self-evaluation, and help seeking.

Some other studies have also explored the relationship between self-regulated strategies and writing performance (e.g. Teng & Zhang, 2024; Zhu et al., 2024). These studies consistently underscore the significant role of SRL behaviors in enhancing writing performance; however, most of these studies rely on static, questionnaire-based data and thus fail to capture the dynamic variations in SRL strategy use among learners.

One exception is the work of Aksela et al. (2025), who analyzed digital trace data and employed network metrics to compare the SRL strategies of higher- and lower-performing students. Their findings revealed that higher-performing students demonstrated a larger proportion of SRL processes during the task and exhibited distinct cyclical patterns. Nevertheless, writing in their study was conducted within the Moodle environment, which lacks the generative, interactive, and adaptive features of GenAI. These distinctive affordances of GenAI may shape the writing process and the associated regulatory mechanisms. Given that SRL is highly context-dependent (Maimaiti & Hew, 2025), how it unfolds in GenAI-assisted writing contexts remains insufficiently understood.

3.3. SRL in GenAI-assisted learning contexts

GenAI effectively supports SRL by providing personalized, interactive feedback that helps learners monitor performance, adjust strategies, and reflect on progress (Xia et al., 2026). Unlike other traditional resources, it dynamically responds to learners' needs, scaffolds complex tasks, and simulates social interaction, supporting engagement across the forethought, performance, and self-reflection phases of SRL (Lan & Zhou, 2025; Xia et al., 2026). SRL in GenAI-supported contexts has increasingly been examined across multiple learning domains, such as language learning (e.g. Pan et al., 2025; Teng et al., 2026), STEM education (e.g. Ma et al., 2026; Ng et al., 2024), and pre-service teacher education (e.g. Xu et al., 2025). For example, Pan et al. (2025) explored how interactive, personalized SRL support delivered through a GenAI chatbot influenced university EFL learners' use of self-regulated reading strategies and their engagement in reading. Ma et al. (2026) examined learners' SRL behaviors in human–GenAI collaborative programming learning environments. These studies highlight the growing scholarly interest in understanding how GenAI supports learners' self-regulation across diverse educational settings.

As an important domain of language learning, writing has also attracted increasing research attention to learners' use of self-regulatory strategies in GenAI-assisted contexts (e.g. Cheng & Cheung, 2026; Fan et al., 2025; Jin et al., 2025; Teng et al., 2026; Tian & Xiang, 2026; Wang et al., 2025). However, existing studies have largely focused on general writing contexts, with relatively little attention paid to academic writing. Academic writing differs from other types of writing in its cognitive and linguistic demands, requiring not only linguistic accuracy but also higher-order skills such as coherent discourse organization, critical thinking, and argumentation (Zhang et al., 2025). In addition, academic writing is typically source-based (Cumming et al., 2016), and learners must integrate information from multiple sources and synthesize them into a coherent text. These demands require learners to actively regulate their strategies during the writing process, making their SRL processes potentially more complex than those involved in other forms of writing. Therefore, research that facilitates a deeper understanding of learners' self-regulatory processes in GenAI-assisted academic writing is particularly important.

4. Methods

4.1. Research design

To investigate how learners engage in SRL during generative AI–assisted academic writing, this study adopted a case study design (Yin, 2009), and employed a process-oriented learning analytics approach. Rather than relying on retrospective self-report data, the study analyzed learners' behavioral trace data collected during authentic writing activities.

Process data were collected through screen recordings of students' writing sessions, capturing their real-time interactions during the writing process. The recordings were subsequently segmented into behavioral units and coded according to an established SRL behavioral coding framework. The coded data were then analyzed using multiple learning analytics methods. First, k-means clustering was applied to explore learner groupings. Second, descriptive statistics and non-parametric tests were conducted to examine differences in

SRL behavior frequencies across clusters. Finally, lag sequential analysis was performed to explore differences in SRL behavioral sequences among learner groups. Similar process-oriented learning analytics approaches have been widely used to investigate learning processes in educational technology research (e.g. Geng & Su, 2025; Ma et al., 2026; Xu et al., 2023).

4.2. Participants

A total of 35 graduate students from a research-oriented university in East Asia initially participated in this study. All were native Chinese speakers and EFL learners who learned English primarily through formal instruction but rarely used it in their daily lives. Participants were recruited voluntarily through course announcements. However, five participants were excluded from the analysis because of incomplete data, including missing screen recordings ($N = 2$) and unavailable English proficiency test scores ($N = 3$). As a result, the final analysis included academic writing data from 30 students (11 males and 19 females; Mean age = 21.48, $SD = 0.69$). Their English proficiency, as indicated by their College English Test Band 6 (CET-6) scores, had an average of 514.8 ($SD = 45.66$). The CET-6 is a nationwide, large-scale English proficiency test in China that assesses advanced skills in listening, reading, writing, and translation. According to Jin et al. (2022), a CET-6 score of 438 corresponds to the minimum threshold of the B2 level in the Common European Framework of Reference for Languages (CEFR), while a score of 596 corresponds to the minimum threshold of the CEFR C1 level. Therefore, the participants' English proficiency in this study was approximately within the B2–C1 range of the CEFR. Prior to the study, ethical approval was obtained, and all participants were informed of the research objectives and procedures and provided written informed consent.

4.3. Procedure

The present study comprised two tasks: (a) completing a background information questionnaire, and (b) engaging in GenAI-assisted academic writing. Figure 1 presents an overview of the study procedure.

First, the participants completed a background information questionnaire, which collected demographic information such as name, gender, and age, as well as their English proficiency test scores. All participant information was anonymized, and their privacy was strictly protected throughout the process, with all information kept confidential.

The GenAI-assisted academic writing task was carried out as part of an eight-week course titled *Educational Research Methods*, with each weekly session lasting three consecutive hours. The course was divided into three phases.

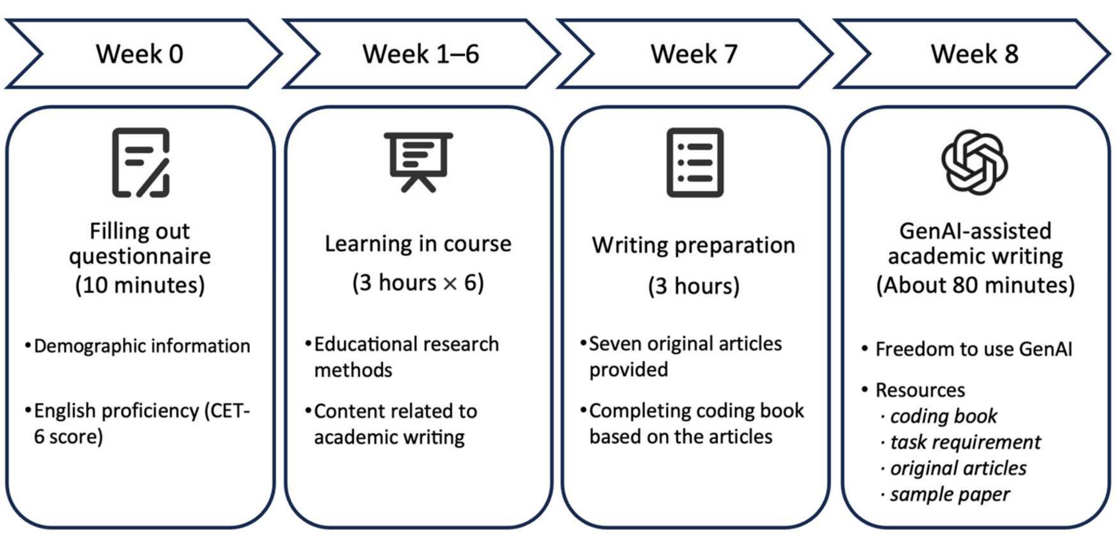

Figure 1. Procedure of the study.

In the first phase (Weeks 1–6), students were introduced to the fundamental methods of educational research, including identifying research questions, research design, and data analysis. Content related to academic writing was also integrated into the course to ensure that students had the necessary foundational knowledge for engaging in academic writing in the final phase.

In the second phase (Week 7), which served as the preparation stage for academic writing, each student was provided with seven research papers on the theme of *gamification in videoconferencing–supported online learning*. Students were required to complete a coding book based on the content of these provided papers. The coding book included sections such as the title, research objectives, methods, findings, innovations, and limitations. This stage primarily aimed to familiarize students with the topic of academic writing.

In the final phase (Week 8), students undertook a formal academic writing task. Students were instructed to use ChatGPT (powered by GPT-3.5) as the GenAI tool during the academic writing task. ChatGPT was selected because it is one of the most widely used and frequently examined GenAI tools in educational research (e.g. Albadarin et al., 2024; Lo, 2023). GPT-3.5 was used as it represented the most up-to-date model available at the time of data collection. Before beginning, the instructor introduced the use of the GenAI tool and distributed the task requirements through an electronic document. Students were asked to compose a literature review of approximately 300 words in English, based on the coding book and seven original research articles provided one week in advance. A sample paper was also provided as a reference to guide students in understanding the expected standard and to assist them in setting appropriate goals for their own writing. During the writing process, students had the freedom to use GenAI and completed their academic writing tasks on their laptops with its assistance. The language of prompts was not restricted, allowing students to draw on all available linguistic resources (e.g. their first language, the target language, or a combination of both), which reflects the authentic conditions under which academic writing with GenAI typically occurs. Students frequently switched between their writing assignment, the

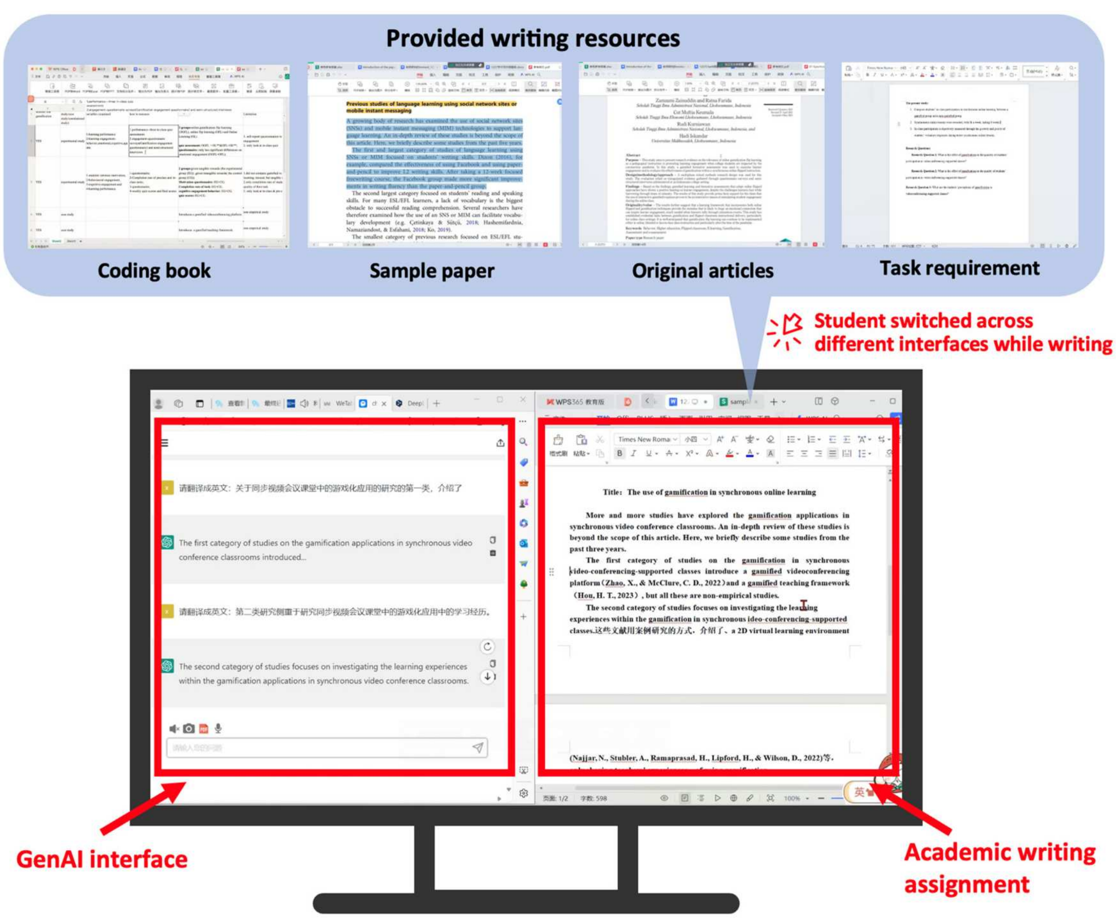

Figure 2. Diagram of GenAI-assisted academic writing task.

GenAI interface, and the provided writing resources (e.g. the coding book, task requirements, original articles, and sample papers) while completing the academic writing task (see Figure 2). Their on-screen activities were recorded throughout the process. Upon completion, students submitted both their screen recordings and their academic writing assignments.

4.4. Data collection and preparation

The data collected in this study included students' English proficiency, academic writing scores, and records of their self-regulatory behaviors.

4.4.1. English proficiency

English proficiency was measured using participants' most recent CET-6 scores, a nationwide standardized English proficiency test in China (Jin et al., 2022). These scores were collected prior to the writing task through an electronic questionnaire and verified using institutional records. A total of 30 valid responses were obtained (*mean* = 514.8, *SD* = 45.66).

4.4.2. Academic writing performance

Each participant submitted their written assignment after completing the GenAI-assisted academic writing task. Two raters, both specializing in the field of education and with experience in publishing academic papers, independently evaluated the assignments. Prior to the scoring process, the raters received training to ensure consistent understanding of the rating rubric and evaluation of sample essays. Each assignment was then independently assessed by two raters using a standardized academic writing rubric adapted from Jacobs et al. (1981). The academic writing rubric, with a maximum score of 100, evaluated the assignments across three dimensions: content, organization, and language (see Table 1 for details). Inter-rater reliability was evaluated using the intraclass correlation coefficient (ICC), which demonstrated a high level of consistency between the two raters (ICC = 0.844, $p < .001$). Finally, the academic writing scores of the 30 participants averaged 77.35 (*SD* = 8.67). The academic writing scores, together with participants' English proficiency levels (CET-6 scores), were collected for subsequent use in clustering students into different groups for later analyses.

4.4.3. Self-regulatory behaviors

Students' behaviors during GenAI-assisted academic writing were recorded using laptop screen capture. Each recording lasted approximately 80 minutes, resulting in a total of about 37 hours of screen-capture data. Based on students' screen recordings of GenAI-assisted academic writing, we coded their self-regulatory behaviors. The coding scheme was developed using a combined inductive and deductive approach – behavioral categories were first inductively derived from the screen recordings and then deductively aligned with the three-phase SRL framework. First, an inductive content analysis was conducted to identify specific categories of students' observable actions in the screen recordings of their writing processes (Drisko & Maschi, 2016). We began by familiarizing ourselves with the recordings and then grouped observable behaviors into distinct action categories based on meaningful behavioral units emerging from the data.

Table 1. Rubric for scoring academic writing assignments.

Dimension	Indicator	Explanation
Content (40 points)	Lead-in (10 points)	The lead-in should be clear, focused, and appropriately related to the topic.
	Critical analysis and argumentation (20 points)	The writing should demonstrate critical engagement with the topic, presenting well-reasoned arguments supported by appropriate evidence.
	Conclusion (10 points)	The conclusion should clearly address the research questions and highlight the main findings.
Organization (40 points)	Organization of related literature (10 points)	The literature sources should be logically organized and thematically grouped, clearly demonstrating the relationships and connections among studies.
	Textual coherence and logic (30 points)	The text should demonstrate clear coherence and logical progression, with smooth transitions that effectively connect ideas and ensure overall readability.
Language (20 points)	Linguistic accuracy (10 points)	The writing should demonstrate linguistic accuracy, with precise word choice, correct grammar, and no spelling or typographical errors.
	Fluency and academic style (10 points)	The writing should be fluent and well-structured, using formal and academic language appropriate for scholarly writing.

For example, when a student typed continuous original sentences in the writing document before switching interfaces, this segment was treated as a single behavioral unit and coded as *Compose Text*. The emerging behavioral categories were iteratively reviewed and refined to ensure conceptual clarity and internal consistency. Prior to formal coding, coders completed a training phase involving pilot coding of a subset of the recordings to establish a shared understanding of category definitions. The coding was initially performed by one author using NVivo software, after which a second author randomly selected 10% of the data for independent coding. This process yielded a Cohen's kappa of 0.865, indicating a high level of inter-rater reliability. Any coding discrepancies were resolved through discussion until consensus was reached.

We then adopted a deductive approach to map the identified behavioral categories onto the SRL framework proposed by Zimmerman (2002). The resulting coding scheme is presented in Table 2. This alignment was guided by theoretical definitions of SRL as well as prior studies that have operationalized SRL processes using observable behavioral indicators (Ma et al., 2026; Shen & Bai, 2024). Details of the coding scheme development are provided in Appendix A. In total, 3268 valid instances of students' SRL behavior were recorded in chronological order to facilitate subsequent lag sequential analysis.

4.5. Data analysis

4.5.1. K-means clustering

We employed k-means clustering to group the participants based on their CET-6 scores and academic writing performance, which corresponded to Research Question 1. During the data preprocessing stage, both variables were standardized prior to clustering. This was done to ensure that the two measures, which were originally on different scales, contributed equally to the clustering process. Standardization prevents variables with larger numerical ranges (e.g. CET-6 scores) from dominating the distance calculations, thereby allowing the clustering algorithm to reflect the true underlying structure of the data rather than scale differences.

Then, the optimal number of clusters was determined using the silhouette score and visual verification. A higher silhouette score indicates that participants within the same cluster are more similar to each other, while being more distinct from those in other clusters, thus reflecting a clearer cluster structure (Rousseeuw, 1987). Silhouette scores were calculated for cluster solutions with k ranging from 2 to 15 (see Figure 3). The solution with three clusters yielded the highest silhouette score (0.472), suggesting relatively clear separation among clusters. Then, following Bai et al. (2025), we conducted a visual verification step. Consistent with the approach of Iatrellis et al. (2021), we visually inspected the clustering solutions with $k = 2$, 3, 4, and 5 (see Figure 4). As shown in Figure 4(a), the clustering solution with $k = 2$ exhibited under-clustering, as indicated by large within-cluster dispersion and the grouping of individuals with differing levels of writing performance into the same cluster. When the number of clusters was increased to four or more, the separation between clusters became less distinct, suggesting potential over-segmentation (see Figure 4(c and d)). By contrast, the three-cluster solution demonstrated relatively clearer separation among clusters and was therefore considered to provide better separability and interpretability with respect to English proficiency and academic writing performance (see Figure 4(b)). Accordingly, the k-means algorithm was applied to classify students into three clusters based on their English proficiency (CET-6 scores) and academic writing performance.

4.5.2. Analysis of SRL behavior frequencies

To address Research Question 2, we quantified the frequency of students' SRL behaviors within each learner cluster based on the established SRL behavior coding scheme (see Table 2). For each SRL phase (forethought, performance, and self-reflection), we calculated the average frequency of coded behaviors per student within each cluster. Given the relatively small sample size and evidence of non-normal distribution, the assumptions required for parametric tests were not met. Therefore, a non-parametric Kruskal–Wallis test was conducted to examine whether there were statistically significant differences in SRL behavior frequencies across clusters. In addition, bar charts were used to visualize the frequencies, providing an exploratory overview of phase-specific SRL behaviors across clusters.

Table 2. Coding scheme for SRL behaviors in GenAI-assisted academic writing process.

SRL Phase	Dimension	Behavioral Code	Description	Example
Forethought phase	Task analysis	Check task requirement (CTR)	Students refer to the electronic document to review the requirements for the current academic writing task.	In Student A's screen recording (5:28.8–8:26.0), the student opened the task requirement document and scrolled up and down to review background information for the current literature review task, including the writing topic, research questions, etc.
		Review original article (ROA)	Students consult the original research papers that were provided.	In Student B's screen recording (2:45.6–8:08.6), the student switched between the provided original articles, moving the cursor over the words and phrases.
		Review coding book (RCB)	Students refer to the provided spreadsheet coding book, which documented key information for each article.	In Student F's screen recording (50:34.9–50:55.7), the student switched the screen to the coding book, remained on this interface for approximately 20 seconds, and then switched away.
		Review sample paper (RSP)	Students refer to the provided sample paper for guidance in their academic writing.	In Student S's screen recording (0:58.9–3:22.1), the student switched the screen to the provided writing sample paper, scrolling up and down.
Performance phase	Self-control	Seek help (SH)	Students asked GenAI for assistance with aspects of their academic writing.	In Student F's screen recording (1:00:01.7–1:00:51.1), the student switched to the GenAI interface, typed the prompt *"What is a self-report questionnaire?"* into the input box, and clicked the send button, after which the GenAI generated several paragraphs in response.
		Compose text (CT)	Students produce text within their academic writing documents.	In Student U's screen recording (1:17:40.3–1:18:56.8), the student typed *"Gamification enhances students' engagement dramatically."* in the academic writing document.
		Copy and paste (CP)	Students perform copy-and-paste actions.	In Student V's screen recording (36:01.4–36:41.7), the student selected the GenAI-generated content *"compare the effects of intangible and tangible rewards on student motivation and engagement"*, then switched to the academic writing document and pasted the text into the draft.
Self-reflection phase	Self-judgement	Revise (R)	Students revise their previously written text during the writing process.	In Student X's screen recording (1:10:37.8–1:10:43.9), the student revised the draft by replacing *"classrooms"* with *"online learning"* in the phrase *"and how to enhance student engagement in synchronous classrooms has triggered thinking"*.

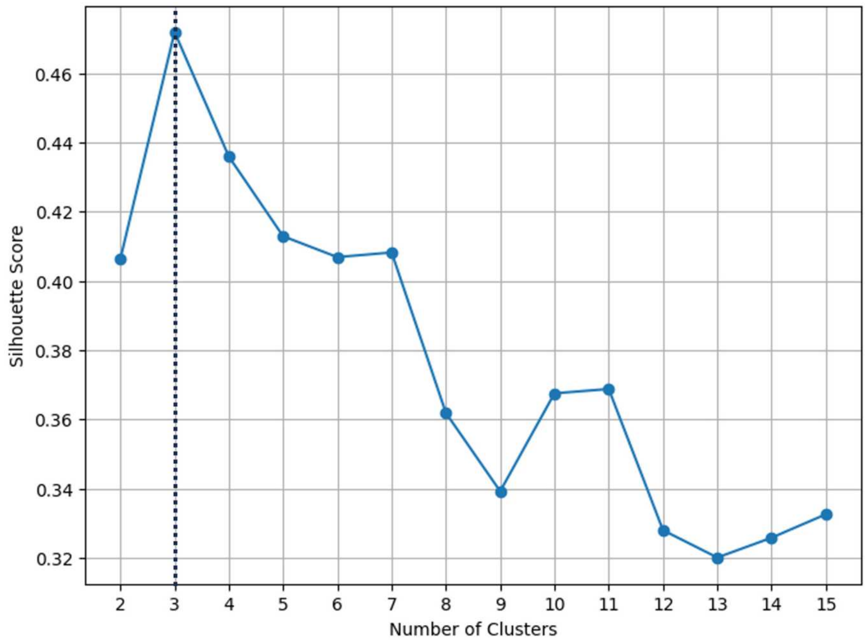

Figure 3. Silhouette scores of different clusters when performing k-means clustering.

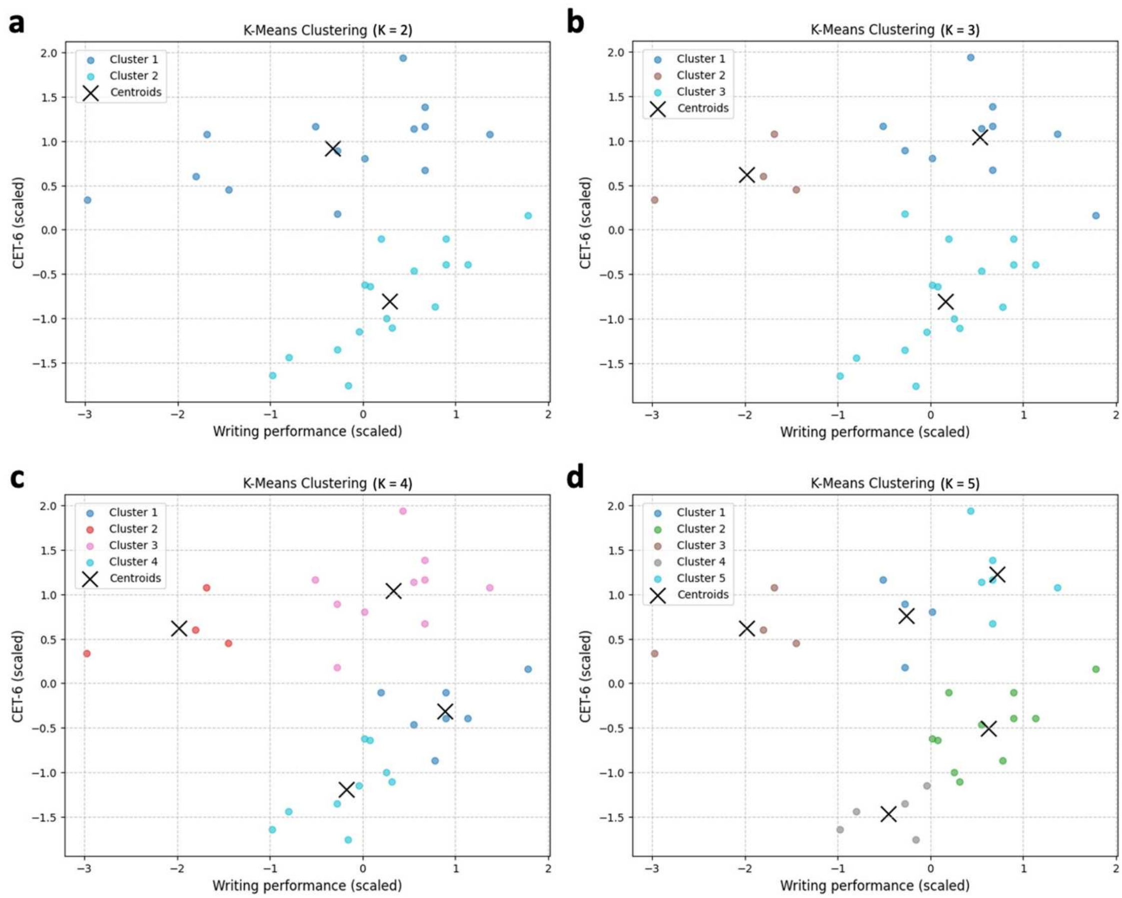

Figure 4. Visual verification of the optimal number of clusters when performing k-means clustering.

4.5.3. Lag sequential analysis

The lag sequential analysis was employed to investigate the differences in SRL behavioral patterns among the groups (Research Question 3). Lag sequential analysis is a widely used method in learning analytics that helps examine the dynamic and temporal nature of interactions. It identifies how one behavior tends to follow another over time, revealing underlying sequential structures in learning processes (Bakeman & Gottman, 1997). Prior to the analysis, students' SRL behaviors were coded based on the established behavioral coding scheme (see Table 2) and arranged chronologically to form individual behavior sequences. In this study, the behavioral sequences of students in each cluster were analyzed using GSEQ software to compute adjusted residuals (z-scores), where a z-score greater than 1.96 indicated a significant sequential relationship between two behaviors. This analysis focused on identifying and comparing significant behavioral transitions across the three groups of students to reveal how they regulated their academic writing processes.

5. Results

5.1. Learner groups based on students' English proficiency and GenAI-assisted writing performance

Students' English proficiency levels and academic writing performance were jointly considered in a k-means clustering analysis. Figure 5 displays the distribution of participants in the two-dimensional space defined by CET-6 scores and academic writing performance, with each cluster indicated by a distinct color. The centroids of each cluster are marked by a black cross and represent the mean values of CET-6 scores and academic writing performance for the cluster.

Three clusters emerged, which were interpreted based on their distributions: (a) students with relatively higher English proficiency but lower academic writing performance (HighEng–LowPerf Group, $N = 4$; mean CET-6 score = 542.50, $SD = 14.53$; mean academic writing score = 60.50, $SD = 5.80$); (b) students with relatively higher English proficiency and higher academic writing performance (HighEng–HighPerf Group, $N = 10$; mean CET-6 score = 561.50, $SD = 20.95$; mean academic writing score = 81.90, $SD = 5.90$); and (c) students with lower English proficiency but higher academic writing performance (LowEng–HighPerf Group, $N = 16$; mean CET-6 score = 478.69, $SD = 26.00$; mean academic writing score = 78.72, $SD = 5.10$). These clusters were subsequently used to examine differences in students' self-regulatory behaviors during the GenAI-assisted writing task.

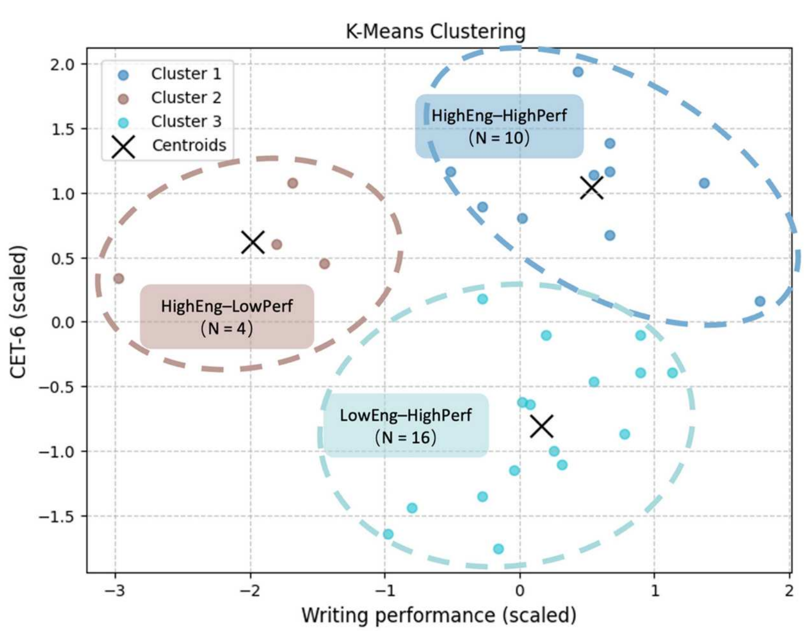

Figure 5. K-means clustering of students based on English proficiency (CET-6 scores) and academic writing performance.

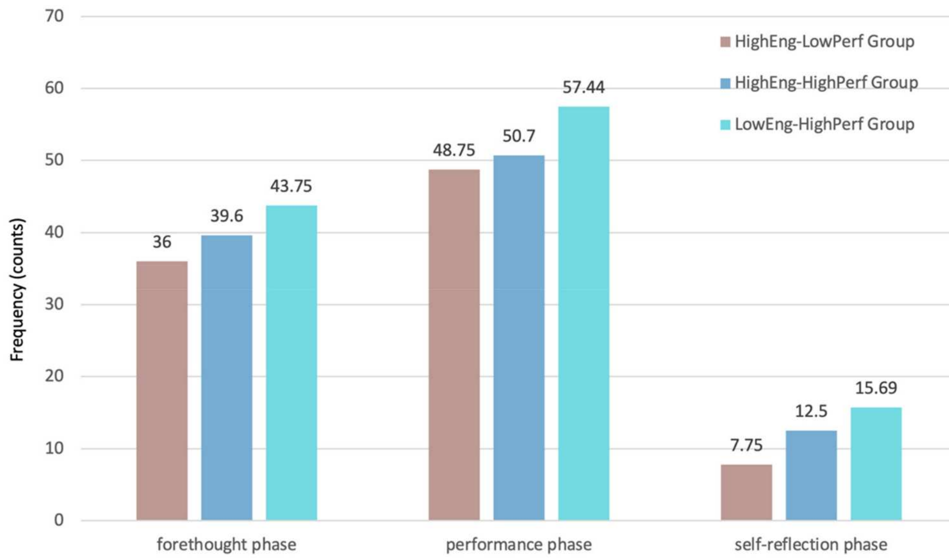

Figure 6. Average frequencies of students' SRL behaviors during GenAI-assisted academic writing process.

5.2. Frequency of SRL behaviors in GenAI-assisted academic writing process

The average frequency of each student's SRL behaviors during GenAI-assisted academic writing was calculated. Figure 6 illustrates the average frequency of SRL behaviors for each group across the three phases, with clusters represented by distinct colors.

A Kruskal–Wallis test indicated no statistically significant differences in SRL behavior frequencies across clusters in the forethought phase ($H(2) = 0.157$, $p = .925$), the performance phase ($H(2) = 0.506$, $p = .776$), or the self-reflection phase ($H(2) = 2.406$, $p = .300$). Descriptive analyses showed that, across the three phases, the HighEng–LowPerf, HighEng–HighPerf, and LowEng–HighPerf groups showed numerically increasing average SRL behavior frequencies, though the differences were not statistically significant. Specifically, in the forethought phase, the averages were 36.00, 39.60, and 43.75; in the performance phase, 48.75, 50.70, and 57.44; and in the self-reflection phase, 7.75, 12.50, and 15.69, respectively.

5.3. SRL behavioral patterns in GenAI-assisted academic writing process

Using students' behavioral sequences, we conducted lag sequential analyses within each cluster to examine differences in behavioral patterns across clusters. A total of 3268 behavioral units were included in the analysis (HighEng–LowPerf Group: 370 behavioral units; HighEng–HighPerf Group: 1028 behavioral units; LowEng–HighPerf Group: 1870 behavioral units). Figure 7 illustrates the sequential patterns of students' SRL behaviors across clusters. Each node represents a specific behavior, with larger nodes indicating higher frequencies of that behavior. The arrows connecting the nodes represent transitions between behaviors, with thicker arrows corresponding to more frequent transitions. Complementing Figure 7, Table 3 reports the summary of key SRL behavioral transitions observed within each cluster.

In the HighEng–LowPerf Group (see Figure 7(a)), a total of 11 significant behavioral sequences were identified. Several key transitions were observed, including frequent switches between writing and referring to original articles (CT → ROA, z-score = 7.14; ROA → CT, z-score = 5.13), as well as recurrent transitions among reviewing the coding book, reviewing sample papers, and checking task requirements (RCB → RSP, z-score = 2.56; RCB → CTR, z-score = 2.55; RSP → RCB, z-score = 2.30; RSP → CTR, z-score = 2.70). Furthermore, students' use of GenAI was primarily associated with behaviors in the forethought phase (RCB → SH, z-score = 3.58; SH → RCB, z-score = 4.11) and performance phase (SH → CP, z-score = 2.29), rather than being directly integrated into the writing process during the revision in the self-reflection phase.

In contrast, the HighEng–HighPerf Group (see Figure 7(b)) exhibited a different behavioral pattern, with 10 significant behavioral sequences. Compared to the HighEng–LowPerf Group, this group of students was more likely to refer to their coding book rather than the original articles while writing (CT → RCB, z-score

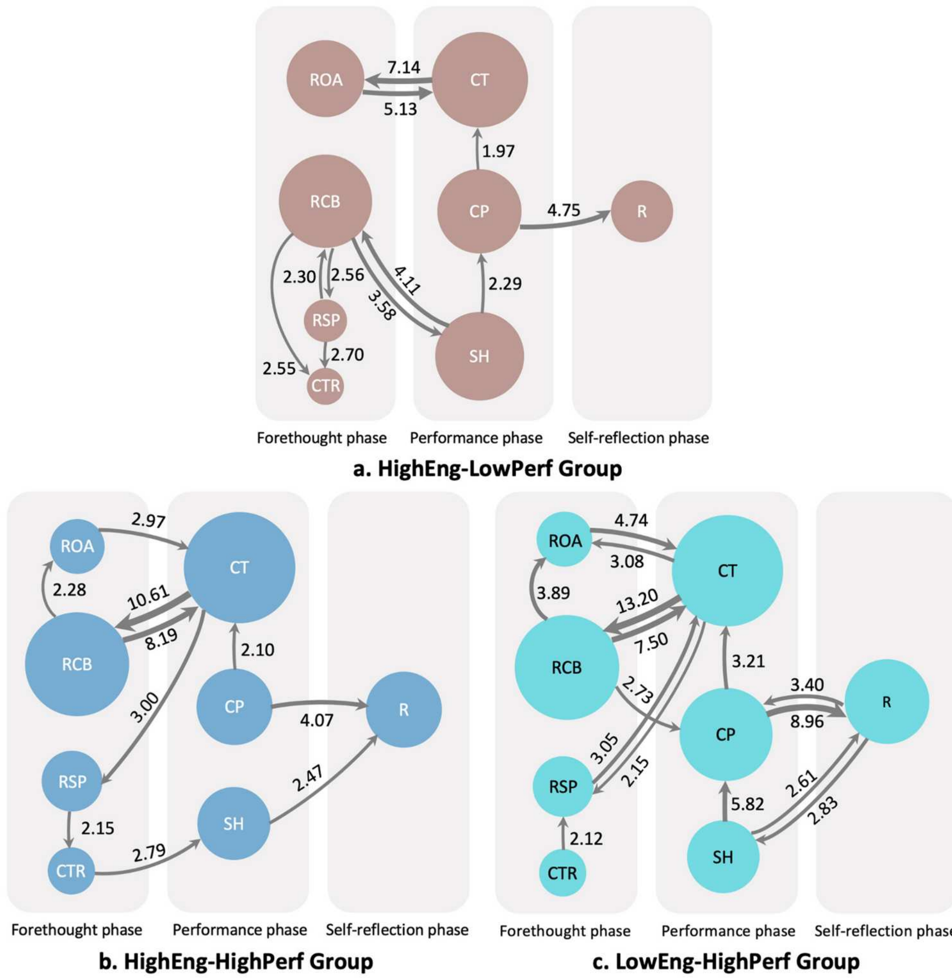

Figure 7. Sequential patterns of SRL behaviors during GenAI-assisted academic writing across student clusters.
Note: ROA: Review original article; RCB: Review coding book; RSP: Review sample paper; CTR: Check task requirements; CT: Compose text; CP: Copy and paste; SH: Seek help; R: Revise.

= 10.61, RCB → CT, z-score = 8.19). They also exhibited a set of recurrent pairwise transitions forming a cyclical configuration, including frequent transitions from writing to reviewing the coding book (CT → RCB, z = 10.61), from reviewing the coding book to reviewing the original articles (RCB → ROA, z = 2.28), and from reviewing the original articles back to writing (ROA → CT, z = 2.97). In addition, GenAI was more frequently used to assist in revising the drafts during the self-reflection phase (SH → R, z-score = 2.47).

Similarly, the LowEng–HighPerf Group (see Figure 7(c)) exhibited a structure generally consistent with that of the HighEng–HighPerf Group, such as the recurrent interaction between writing and behaviors in the forethought phase (CT → RCB, z-score = 13.20; RCB → ROA, z-score = 3.89; ROA → CT, z-score = 4.74), and the use of GenAI to support the revision process (SH → R, z-score = 2.61). In addition, this group exhibited numerous reciprocal transitions across behaviors, including repeated back-and-forth movements between writing and reading original articles (ROA → CT, z-score = 4.74; CT → ROA, z-score = 3.08), between copying–pasting and revising (CP → R, z-score = 8.96; R → CP, z-score = 3.40), and between querying GenAI and revising (SH → R, z-score = 2.61; R → SH, z-score = 2.83). Notably, this group also exhibited transitions from querying GenAI to copying–pasting text (SH → CP, z-score = 5.82). Overall, the present group exhibited a more complex behavioral network, characterized by 15 significant sequences and multiple reciprocal transitions among SRL phases.

6. Discussion

The present study collected students' English proficiency scores (CET-6), screen-recorded data from their GenAI-assisted learning sessions, and their final writing performance. First, k-means clustering was

Table 3. Summary of SRL behavioral patterns within each cluster.

Cluster	Pattern description	Key transitions (z-score)
HighEng–LowPerf Group	Learners frequently switched between composing text (CT) and referring to original articles (ROA).	CT → ROA (7.14), ROA → CT (5.13).
	Learners exhibited recurrent transitions among reviewing the coding book (RCB), reviewing sample papers (RSP), and checking task requirements (CTR) within the forethought phase.	RCB → RSP (2.56), RCB → CTR (2.55), RSP → RCB (2.30), RSP → CTR (2.70).
	Learners primarily used GenAI (SH) to support their understanding of the coding book (RCB) and also exhibited frequent transitions from querying GenAI to copy-and-paste actions (CP).	RCB → SH (3.58), SH → RCB (4.11), SH → CP (2.29).
HighEng–HighPerf Group	Learners were more likely to refer to the coding book (RCB) during composing (CT) and exhibited recurrent pairwise transitions among composing (CT), reviewing the coding book (RCB), and consulting original articles (ROA), forming a cyclical configuration.	RCB → CT (8.19), CT → RCB (10.61), RCB → ROA (2.28), ROA → CT (2.97).
	Learners primarily used GenAI (SH) for revising their drafts (R).	SH → R (2.47).
LowEng–HighPerf Group	Learners tended to refer to the coding book (RCB) while composing text (CT) and showed recurrent pairwise transitions among composing (CT), reviewing the coding book (RCB), and consulting original articles (ROA), indicating a cyclical pattern of engagement across these activities.	RCB → CT (7.50), CT → RCB (13.20), RCB → ROA (3.89), ROA → CT (4.74).
	Learners' SRL behaviors exhibited numerous reciprocal transitions across SRL phases.	ROA → CT (4.74), CT → ROA (3.08); RCB → CT (7.50), CT → RCB (13.20); CP → R (8.96), R → CP (3.40); SH → R (2.61), R → SH (2.83).
	Learners mainly used GenAI (SH) to assist with revising their drafts (R) and demonstrated transitions from querying GenAI to copy-and-paste actions (CP).	SH → R (2.61), R → SH (2.83), SH → CP (5.82).

applied using English proficiency and writing performance as key indicators to explore distinct learner clusters. Next, students' SRL behaviors within each cluster were analyzed by calculating average frequencies and conducting lag sequential analysis, which revealed different behavioral patterns. In this section, we further discuss these findings in detail.

6.1. Characteristics of learner clusters

The clustering analysis suggested three clusters based on students' English proficiency and writing performance: (1) students with higher English proficiency but lower writing performance (HighEng–LowPerf Group, $N = 4$); (2) students with higher English proficiency and higher writing performance (HighEng–HighPerf Group, $N = 10$); (3) students with lower English proficiency but higher writing performance (LowEng–HighPerf Group, $N = 16$). This clustering pattern is broadly in line with Aksela et al. (2025), who also categorized students based on high and low writing performance. By incorporating English proficiency as an additional indicator in the clustering process, the present study adopts a multidimensional perspective that may provide a more nuanced view of learners' characteristic profiles.

Interestingly, the results revealed a distinctive clustering pattern, particularly with the appearance of the HighEng–LowPerf Group and the LowEng–HighPerf Group. This finding is different from the commonly held view that English proficiency tends to be positively associated with writing performance (Lahuerta Martínez, 2024; Sasaki & Hirose, 1996; Wu et al., 2021). The occurrence of such a pattern may be explained by the affordances of GenAI-supported writing environments. In traditional, teacher-centered writing classes, where teachers control judgments about the legitimacy of information and what constitutes knowledge (Kain, 2003), students typically have limited opportunities to actively employ their learning strategies that could improve their writing performance. By providing interactive and personalized feedback, GenAI offers students more opportunities to continuously adjust their learning strategies during the writing process (Chang & Sun, 2024; Lan & Zhou, 2025; Xia et al., 2026), which may thereby lead to greater variability in their writing outcomes depending on how effectively they regulate their learning. These findings highlight the necessity of further examining SRL behaviors across different clusters of students, which may shed light on the divergent writing outcomes emerging in GenAI-supported environments.

6.2. Frequencies of SRL behaviors across clusters

Unlike previous studies that primarily examined variations in students' SRL frequency across varying cognitive load (e.g. high- vs. low-cognitive-load tasks) or different feedback sources (Cheng & Cheung, 2026; Fan et al., 2025), the present study situates SRL frequencies within learner profiles characterized by combinations of English proficiency and writing performance. This cluster-based perspective offers a more learner-centered perspective on how SRL engagement relates to writing outcomes in GenAI-assisted contexts.

Although no statistically significant differences were found in SRL behavior frequencies across clusters, the descriptive trends provide some insights into learners' self-regulatory engagement. Specifically, the LowEng–HighPerf group tended to exhibit relatively higher average frequencies of SRL behaviors across the forethought, performance, and self-reflection phases, followed by the HighEng–HighPerf and HighEng–LowPerf groups. This pattern suggests that more frequent engagement in self-regulatory activities may be associated with more effective writing performance, and that this association may be less strongly influenced by learners' initial English proficiency, consistent with prior research (Teng & Zhang, 2024). Given the exploratory nature of the study, these findings should be interpreted as preliminary trends that require further validation in future research.

6.3. SRL behavioral patterns across clusters

To explore the behavioral patterns across the three different clusters, we conducted lag sequential analysis based on students' SRL behaviors.

6.3.1. Comparison in higher- and lower-performing groups

By comparing the behavioral sequential patterns of the lower-performing group (i.e. the HighEng–LowPerf Group) with those of the higher-performing groups (i.e. the HighEng–HighPerf Group and the LowEng–HighPerf Group), clear differences in SRL behaviors were identified. Although the HighEng–LowPerf Group was small in size, its distinct behavioral pattern provides exploratory insights.

On the one hand, students in the higher-performing groups exhibited a higher level of self-regulation during GenAI-assisted academic writing, as evidenced by a relatively large number of cyclical behavioral sequences (see Figure 7(b,c)). This behavioral pattern is consistent with the cyclical characteristic described in Zimmerman's (2002) model of self-regulation. In contrast, students in the lower-performing group, despite possessing strong language proficiency, displayed a lower level of self-regulation, characterized by fewer and relatively uncoordinated transitions between SRL phases (see Figure 7(a)). This finding aligns with previous research (Aksela et al., 2025) and further underscores the important role of SRL strategies in GenAI-assisted writing contexts.

On the other hand, clear differences in SRL behavioral sequence transitions were observed between the lower- and higher-performing groups (see Figure 7). For instance, students in the higher-performing groups frequently exhibited cyclical transitions between the forethought and performance phases, and they were also more likely to transition smoothly from planning to textual production. Specifically, they engaged in recurrent adjacent transitions among reviewing the coding book to extract key information, consulting the original article to verify details, and transforming the information into text (RCB → CT; CT → RCB; RCB → ROA; ROA → CT) (see Figure 7(b,c)). In contrast, students in the lower-performing group tended to repeatedly read the provided writing materials under GenAI assistance (with action sequences frequently switching among RCB, RSP, CTR, and SH), reflecting a disconnect between their writing planning and actual writing execution (see Figure 7(a)). This finding aligns with previous research (Aksela et al., 2025) and may be attributed to differences in students' awareness and effective use of metacognitive strategies: higher-performing students can better monitor, regulate, and align their writing plans with execution, whereas lower-performing EFL writers often show a lack of awareness of the importance of metacognitive strategies and are unable to employ them effectively (Hosseinpur & Kazemi, 2022).

Compared with the GenAI use observed in the lower-performing group, the higher-performing groups exhibited behavioral sequences between seeking assistance from GenAI and revising their articles (SH → R), indicating that students actively engaged in self-evaluation and reflection with AI support (see Figure 7(c)). This can be explained by the structural model of AI-assisted writing (Jin et al., 2025), which indicates that different levels of AI use have distinct effects on writing outcomes. In particular, using AI for text development (e.g. content revision) has a significant positive impact on writing improvement (Jin et al., 2025). Notably, draft revision behaviors have been frequently observed in GenAI-assisted writing contexts, including both the present study and prior research (Fan et al., 2025). In contrast, such behaviors were not reported in a Moodle-based writing environment without GenAI support (Aksela et al., 2025), possibly due to the limited scaffolding available to support students' self-evaluation on that platform. From a sociocultural perspective, GenAI's feedback provides learners with a form of cognitive scaffolding (Liu & Zhang, 2025; Su et al., 2023), facilitating monitoring and adjustment of their drafts. In this context, learners can use GenAI to continuously refine their drafts during the self-reflection phase, which may contribute to improvements in the quality of their writing.

6.3.2. Comparison in higher- and lower-English proficiency groups within higher performers

Within the higher-performing students, both the HighEng–HighPerf and LowEng–HighPerf groups exhibited similar overall behavioral patterns (see Figure 7(b,c)). However, despite achieving comparable final writing scores, the HighEng–HighPerf Group demonstrated a noticeably smaller range of SRL behavioral sequences (see Figure 7(b)). From a cognitive psychology perspective, higher English proficiency is generally associated with greater working memory capacity (Linck et al., 2014), enabling learners to allocate cognitive resources more efficiently during writing; consequently, these learners may achieve strong writing performance with relatively fewer overt SRL behaviors.

By contrast, the LowEng–HighPerf Group exhibited a denser and more interconnected set of SRL behavioral sequences (see Figure 7(c)), characterized by frequent reciprocal behaviors through which learners continuously regulated their learning, with GenAI serving as a supportive resource in the iterative refinement of their written products (SH → R; R → SH). This pattern suggests that, in GenAI-assisted writing contexts, these students may have engaged in more dynamic self-regulation, potentially as a way to offset challenges associated with lower language proficiency. This finding is consistent with prior research showing that students achieve better writing outcomes when engaging in iterative, highly interactive processes supported by GenAI (Nguyen et al., 2024), and further suggests that SRL strategies have a stronger effect on learners with lower English proficiency (Teng & Zhang, 2024). It also aligns with Zimmerman's (2002) theoretical claim that students' self-regulation can function as a means of compensating individual differences in learning. This group may be interpreted as reflecting a shift from a deficit-oriented view of lower-English-proficiency learners to a regulation-oriented perspective, in which strategic engagement with AI supports learners in managing language-related challenges, thereby offering important implications for equity-oriented AI pedagogy in EFL contexts. Future research could construct more fine-grained learner profiles by incorporating a broader range of learner characteristics to further investigate the mechanisms through which SRL behavioral patterns compensate for individual differences.

However, the SRL behavioral sequences of this group also revealed potential drawbacks, as students frequently copied and pasted GenAI-generated content (SH → CP). This finding suggests that the integration of GenAI into academic writing may entail both opportunities and challenges. Consistent with prior research, while GenAI can reduce cognitive load during learning processes (Stadler et al., 2024), uncritical use may lead to potential overreliance on GenAI (Chan & Lee, 2025). Therefore, a nuanced understanding of how to leverage the strengths of GenAI while mitigating its limitations could inform future research on GenAI-assisted academic writing.

7. Implications

This section presents implications of our findings from both theoretical and pedagogical perspectives.

7.1. Theoretical implications

The present study was grounded in Zimmerman's (2002) cyclical model of self-regulation and situated this framework within the context of GenAI-assisted academic writing by examining learners' dynamic SRL behaviors during AI-supported writing. This study provides a more nuanced understanding of how self-regulatory processes unfold in technology-mediated writing environments.

Furthermore, we proposed an adapted coding framework for SRL behaviors – structured around the forethought, performance, and self-reflection phases – offering a theoretically informed analytical tool for capturing SRL processes in GenAI-assisted academic writing.

By incorporating both English proficiency and final writing performance as key learner characteristics, this study provides initial empirical support for the possibility that effective SRL strategies may help mitigate limitations in linguistic ability in GenAI-assisted academic writing contexts. This interpretation is in line with theoretical claims regarding the compensatory role of self-regulation in learning (Zimmerman, 2002). It is best understood as providing context-specific and exploratory insights rather than definitive theoretical validation. Future research could include additional learner characteristics (e.g. gender, years of English learning experience, interest in English writing, etc.) to further unpack the mechanisms through which SRL strategies interact with individual differences in GenAI-supported learning contexts.

7.2. Pedagogical implications

Based on the findings, the present study proposes the following pedagogical implications for GenAI-assisted academic writing. These implications suggest that EFL writing instruction may benefit from moving beyond mere language correction toward an increased emphasis on strategic regulation training.

First, writing instruction should prioritize developing students' AI literacy to enhance their self-regulatory strategies during the GenAI-assisted writing process (Shi et al., 2025). Learners need not only to master the basic operation of GenAI tools, but also to strategically regulate their interactions with GenAI, such as planning prompts, monitoring feedback, and critically assessing AI-generated writing suggestions. Instructors can incorporate workshops on the strategic use of GenAI tools into writing courses to build students' AI literacy, thereby supporting and strengthening their self-regulatory behaviors.

Second, instructional design should be differentiated according to learners' individual characteristics, such as learners' self-regulatory profiles. The findings showed that lower-performing learners, despite relatively higher English proficiency, exhibited weaker integration between planning and writing execution. For these learners, instructors can provide structured SRL scaffolds – such as phase-specific prompts, planning templates, and reflective checklists – to strengthen the connection between forethought and performance. In contrast, higher-performing learners demonstrated more cyclical and integrated SRL patterns and may benefit from greater autonomy and opportunities for critical engagement with GenAI during drafting and revision.

Lastly, writing instruction should explicitly address the risk of overreliance on GenAI (Chan & Lee, 2025). The findings indicated that frequent copying of AI-generated content may coexist with strong performance, particularly among learners with lower English proficiency. To mitigate this risk, instructors should frame GenAI as a cognitive partner rather than a substitute for learners' thinking and encourage reflective practices, such as requiring students to justify how AI feedback is adopted or revised, to maintain critical engagement and ethical academic writing.

8. Limitations

Although this study uncovers the dynamic SRL behaviors of learners across different clusters in GenAI-assisted academic writing contexts, several limitations should be acknowledged and addressed in future research.

First, this study was conducted with a relatively small sample size, which constrained its generalizability. As a result, the learner clusters and frequency distributions should be interpreted as exploratory rather than exhaustive. Future research could involve larger and more diverse samples across institutions and instructional contexts to validate the robustness of the clustering results and to capture a broader range of learner profiles and self-regulatory patterns.

Second, this study focused on a single academic writing task of a specific genre, namely, a literature review. However, SRL behaviors are highly context-dependent (Maimaiti & Hew, 2025), and different types of writing tasks (e.g. research protocol writing, academic argumentative writing) may elicit different SRL patterns. Future research should therefore incorporate a wider range of tasks and genres to enhance the external validity of the findings.

Third, the motivational processes underlying learners' selection and adjustment of SRL strategies, which may not be externally observable, were not captured. Future studies could benefit from physiological data – such as EEG and eye-tracking data – or by incorporating qualitative methods, including stimulated recall interviews or think-aloud protocols, to gain deeper insights into why learners make strategic decisions and how they regulate their learning during GenAI-assisted writing.

9. Conclusion

The emergence of GenAI has provided students with enhanced linguistic support and expanded opportunities to engage in self-regulatory strategies. However, limited research has examined how students with different English proficiency levels dynamically enact SRL and how this relates to their academic writing outcomes in GenAI-assisted contexts. To address this gap, this study analyzed the GenAI-assisted academic writing of 30 graduate students using multiple learning analytics methods, including clustering, descriptive statistics, and lag sequential analysis. Specifically, we first clustered the students by jointly considering their English proficiency and writing performance. The analysis suggested three possible cluster patterns among the participants: HighEng–LowPerf, HighEng–HighPerf, and LowEng–HighPerf. Subsequently, we conducted lag sequential analyses on a total of 3268 behavioral units within each cluster. The results indicate that, in GenAI-assisted academic writing contexts, higher-performing students tended to exhibit more cyclical and well-coordinated SRL behavioral patterns, whereas lower-performing students showed comparatively fragmented and less coordinated SRL behaviors. Moreover, with the compensatory scaffolding provided by GenAI, some students with lower English proficiency were able to achieve higher writing performance by engaging more actively in self-regulated learning processes. At the same time, the findings also point to a potential risk of uncritical overreliance on GenAI. This study provides a more nuanced understanding of how self-regulatory processes unfold in GenAI-mediated writing environments by proposing a theoretically informed coding framework of SRL behaviors, and offers strategic recommendations for future GenAI-assisted academic writing instruction.

Disclosure statement

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Declaration of generative AI use

ChatGPT-4o was used for enhancing clarity of expression during the manuscript preparation process. The authors have carefully reviewed the manuscript and take full responsibility for all content presented in this article.

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Appendix A. Mapping of observed behaviors to SRL phases in GenAI-assisted academic writing

SRL phase	Dimension	Definition (Zimmerman, 2002)	SRL behavior in GenAI-assisted writing in present study	SRL behavior in writing (Shen & Bai, 2024)	SRL behavior in programming (Ma et al., 2026)
Forethought	Task analysis	Strategic planning prior to task execution	Check task requirement Review original article Review coding book Review sample paper	Planning	Assignment understand Acquire resources
Performance	Self-control	Deployment of strategies during task execution	Compose text Copy and paste Seek help	Text-generating Help-seeking	Code Run
Self-reflection	Self-judgement	Comparisons between one's performance and relevant standards	Revise	Self-evaluation	N/A